DEEPAK KATUKURI

AI ENGINEER · GENAI & AGENTS · MCP



CONTACT

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TECHNICAL SKILLS

GenAI & Agents: LLM APIs (Claude / Anthropic API, Batches API), prompt engineering, structured outputs (JSON schema), LangGraph, LangChain, agentic workflows, human-in-the-loop review, Model Context Protocol (MCP), RAG, Corrective RAG, Chroma, pgvector, embeddings, LLM evaluation

AI Tools: Claude Code (custom MCP servers), Ollama, Hugging Face

Languages: Python, JavaScript, TypeScript, SQL, PL/SQL, Java

APIs & Automation: REST API integration, httpx, FastAPI, Typer CLI, asyncio, scheduled jobs, SQLite, pytest

Full-Stack & Cloud: React, Next.js, Vite, Tailwind CSS, PostgreSQL, Docker, AWS (EC2, RDS), Kubernetes, GitHub Actions, CI/CD, Vercel, Render

ML & Data: PyTorch, Scikit-Learn, Pandas, NumPy, Matplotlib

Tools: Git/GitHub, Jupyter, Linux (Ubuntu)

EDUCATION

MSc Data Science
University of Europe, Germany
Sep 2025 – Present (full-time)

BTech in Computer Science & Engineering
MLRIT, India
Jun 2018 – Jul 2022

CERTIFICATIONS

PL/SQL (Database Management)
Core Java (Software Development)
OneShield (P&C Insurance)

LANGUAGES

English – Fluent
German – A1/A2 (learning)

PROFILE

AI Engineer and Data Science MSc candidate with 2.5 years of production software engineering. Builds prototypes first, then measures: a 16-node LangGraph agent workflow on the Claude API, a TypeScript MCP server that connects the Apify API to Claude Code, and a corrective RAG loop that refuses when it cannot answer. Turned insurance underwriting and claims requirements into technical specs and supported production systems for two client platforms at ValueMomentum. Looking for an English-speaking working student (Werkstudent) role in applied AI and GenAI workflow automation in Germany.

PROJECTS

JobPilot: Multi-Agent Job Search & CV Tailoring Workflow

Python · LangGraph · Claude API · REST APIs · github.com/Deepak589/Job_hunt_Agentic_AI

- Built a 16-node LangGraph agent workflow on the Claude API that pulls jobs from 7 boards via REST APIs, tailors the CV, and blocks any claim without evidence.
- Added human-in-the-loop review, crash recovery and cost tracking; tested upgrades with structured experiments; 238 pytest tests.

Apify MCP Server for Claude Code

TypeScript · Model Context Protocol · Apify API · Claude Code

- Built a TypeScript MCP server (15 tools) that connects the Apify scraping API to Claude Code, so scraping jobs run straight from a chat session.

RAG Pipeline: Hybrid Retrieval to Corrective RAG

Python · LangChain · pgvector · Claude API · github.com/Deepak589/RAG_pipeline

- Built a hybrid RAG retriever with a corrective loop where Claude grades results and refuses unanswerable questions: 100% correct refusal, 69.5% end-to-end correctness.

RA Nutrition & Lifestyle App: Shipped Full-Stack Product

React · JavaScript · FastAPI · PostgreSQL · live at ra-project-indol.vercel.app

- Shipped a live health app that turns daily meal and symptom logs into recommendations and a weekly dashboard, backed by 99 tests.

EXPERIENCE

Software Engineer — Mar 2022 – Sep 2024

ValueMomentum · clients: Hiscox, Philadelphia Insurance

- Resolved 100+ production defects through hands-on troubleshooting across two carrier platforms, Hiscox (P&C insurance) and Philadelphia Insurance (surety bonds). Turned business requirements for underwriting and claims into technical specs, and ran QA/UAT test cases and CI/CD deployments across the full policy lifecycle.
- Migrated full policy, claims, premium and customer history for Philadelphia Insurance's Surety Bonds line from a legacy policy admin system into OneShield PAS. Wrote and validated the PL/SQL migration scripts across customer, policy, claims and premium tables.
- Root-caused a silent data-integrity defect that loaded claims for pre-2019 Surety bonds empty whenever two bonds shared a bond type under different bond IDs, and kept records current with a weekly backfill script while a permanent fix was pending.

ML Intern — Jun – Jul 2019

SmartBridge × IBM

- Built a lung-cancer risk-prediction ensemble (Random Forest, KNN, Gradient Boosting, Neural Network) and deployed it on IBM Watson ML. Presented the results to non-technical reviewers with Matplotlib and Seaborn charts.